POME Pipeline Documentation

Pipeline for Observation of Mid-infrared with ERIS

1 Introduction

POME (Pipeline for Observation of Mid-infrared with ERIS) is a Python-based data reduction pipeline developed to process mid-infrared observations acquired with the **ERIS/NIX** instrument at the VLT. It is specifically tailored for observations with ERIS, as described in the survey presented in *Protoplanetary disk insights from the first ERIS/vAPP survey at 4 μm.*

This document provides a detailed overview of the file structure, script usage, reduction steps, and output products.

2 Image Reduction Procedure

The main reduction steps are:

1. **Master Dark Creation:** A master dark frame is produced by averaging five individual dark frames taken with the same detector integration time (DIT) as the science frames, to minimize thermal noise and mitigate temperature variations.

2. **Flat Field Normalization:** Flat field frames are obtained using twilight sky exposures. Each flat frame is dark-subtracted and then divided by its median value to normalize the response. The final master flat is generated as the median of these normalized frames.

3. **Bad Pixel Correction:** The correction is performed in two stages:

   - A static bad pixel map is provided by ESO based on internal lamp exposures.
   - A dynamic bad pixel map is generated by fitting the image histogram with a Gaussian and identifying pixels beyond 5σ as defective.

   The union of the two maps is used to interpolate and replace the bad pixels in all frames. Large clusters of damaged pixels, however, are masked and excluded from further processing.

4. **Background Subtraction:** Background levels are estimated and subtracted using either sky frames or adjacent nod positions, depending on the observational setup.

5. **Star Alignment:** Images are aligned based on the centroid of the stellar PSF.

6. **Frame Selection:** Individual frames are inspected for quality based on flux level, PSF shape, and background uniformity. Low-quality frames are discarded.

7. **PSF Subtraction:** The final step involves subtracting a reference PSF from each science image to enhance the detectability of faint companions or disk structures. This can be done using angular differential imaging (ADI) or similar algorithms depending on the available data.

3 Directory Structure

The pipeline expects a strict folder hierarchy as follows:

```
<base_dir>/

 raw/                    # Raw FITS files
    <target_name>/       # E.g. HD100546, RYTau

        calib/                 # Calibration files
           dark_flat/          # Darks for flats
           dark_obj/           # Darks for science images
           dark_psf/           # Darks for PSF frames
           flat/               # Flat fields
        object/                # Science cube
        sky/                   # Sky background cube
        psf/                   # Non-coronagraphic PSF images
 reduced/                      # Final processed outputs
```

This structure supports the presence of multiple targets observed with different techniques (e.g., vortex, vAPP), as long as each one is placed in a distinct subfolder under `raw/` with a unique name.

Important notes:

- **Do not rename the FITS files provided by ESO.** The filenames must be preserved exactly as downloaded from the ESO archive to ensure the pipeline can correctly recognize and sort them based on headers and metadata.
- The only exceptions are in the `psf/` folder:
  - `psf_obj.fits` should be the renamed PSF frame associated with the science target.
  - `psf_sky.fits` should be the renamed PSF frame associated with the sky background.

4 Observation Lists for Vortex Mode

When running the pipeline in `vortex` mode, two additional text files must be created manually inside the corresponding target folder under `raw/<target_name>/`:

- `obj_list.txt` — Contains the list of object frames to be processed.
- `sky_list.txt` — Contains the list of corresponding sky frames to be subtracted.

Important notes:

- The files must list the corresponding FITS filenames (without extensions) in chronological order.
- The entries in `obj_list.txt` and `sky_list.txt` must appear in the **same order**, such that:
  - The first object frame in `obj_list.txt` will be paired with the first sky frame in `sky_list.txt`,
  - The second object with the second sky, and so on.
- This pairing ensures a consistent background subtraction per nod position.

5 Observation List for vAPP Mode

When running the pipeline in `vAPP` mode, a single plain text file named `list.txt` must be provided in the corresponding target folder (e.g., under `raw/<target_name>/`). This file should contain one row per A-B pair of observations. Each row must include six columns, separated by spaces or tabs, in the following order:

1. Filename of the science frame at APP position A
2. Filename of the science frame at APP position B
3. X-coordinate of the vAPP PSF center in position A
4. Y-coordinate of the vAPP PSF center in position A
5. X-coordinate of the vAPP PSF center in position B
6. Y-coordinate of the vAPP PSF center in position B

This file allows the pipeline to correctly associate APP pairs and align the PSFs based on the provided centroids.

6 Observation List for Imaging Mode

When running the pipeline in `imaging` mode, a single plain text file named `list.txt` must be provided in the corresponding target folder (e.g., under `raw/<target_name>/`). This file should contain one row per A-B pair of observations. Each row must include six columns, separated by spaces or tabs, in the following order:

1. Filename of the science frame at position A
2. Filename of the science frame at position B
3. X-coordinate of the PSF center in position A
4. Y-coordinate of the PSF center in position A
5. X-coordinate of the PSF center in position B
6. Y-coordinate of the PSF center in position B

These coordinates represent the estimated center of the PSF in each frame, and are used by the pipeline to align the A-B pairs during image processing.

7 Running POME

To launch the full pipeline:

```
python3 pome.py <target_name> <mode>
```

Arguments:

- `<target_name>` — Name of the target directory
- `<mode>` — Observing mode:
  - `vortex` : Supported

- `vAPP` : Not yet supported
- `img_pupil` : Not yet supported
- `img_field` : Not yet supported

Example:

```
python3 pome.py HD100546 vortex
```

8 Running Individual Scripts

Scripts can be run independently for debugging or step-by-step reduction:

```
python3 <script_name>.py <base_path> <target_name>
```

Example:

```
python3 eris_elab_Obj_w3.py /home/data HD100546
```

9 Script Outputs

The following table summarizes each script's output and diagnostic plots.

Script	Main Output	Plots / Diagnostics
`eris_calib_dark_flat_w4.py`	Master dark (flat)	–
`eris_calib_flat_w4.py`	Normalized flat	Histogram
`eris_calib_dark_w4.py`	Master dark (science)	–
`eris_calib_dark_psf_w4.py`	Master dark (PSF)	–
`eris_elab_Obj_w3.py`	Cleaned science cube	Before/after images
`eris_elab_sky_w3.py`	Cleaned sky cube	Sky map
`eris_elab_psf.py`	Cleaned PSF cube	PSF check
`vapp_eris_elab_wf.py`	Residual wavefront maps	PSF residuals
`vapp_eris_background.py`	Background estimation	Background map
`vapp_eris_dh_estraction.py`	Double-lobe PSF separation	DH profile
`save_cube.py`	Final FITS cubes	–

10 Output Directory Structure

All final outputs are saved in a new directory named `reduced/`, located at the same hierarchical level as the `raw/` folder. Inside `reduced/`, a subdirectory is created for each target, using the target name as the folder name (e.g., `HD100546/`).

Each target-specific folder contains the following subdirectories, grouping the output products by type:

```
reduced/
 HD100546/
     calib/      # Calibration frames (master darks, master flat, etc.)
     image/      # Images for diagnostic purposes
     object/     # Final reduced science cubes
     psf/        # Non-coronagraphic PSF images
     sky/        # Master sky observations and background estimates
```

Acknowledgements

This pipeline was developed as part of the ERIS/VLT observing programs focused on the detection and characterization of young planets and circumstellar disks. The methodology and application are described in detail in the paper:

"Protoplanetary disk insights from the first ERIS/vAPP survey at 4 μm", **F. Maio et al.**, 2025.